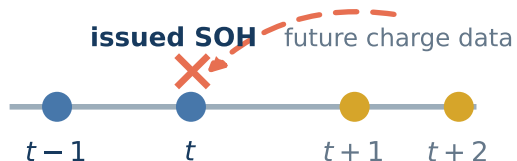


1 Operational SOH gap

Partial charge records arrive first

reference capacity labels arrive later

retrospective revision



Future observations must not rewrite
an already issued health record

2 Prefix-causal projection

with an exact harm budget

Protected state
prefix-only b_t

Learned candidate
record-specific c_t

Exact online viability set

$[b_t - \delta, b_t + \delta] \cap \mathcal{Y} \cap (-\infty, p_{t-1}]$

project candidate to the closest viable output

output p_t

Every hidden outcome: $|p_t - y| - |b_t - y| \leq \delta$

3 Audited evidence

Nested cross-domain

12 dataset domains

11/12 improved | $\Delta\text{MAE} -0.00486$

Outcome-blind external confirmation

45 held-out eligible cells

45/45 improved | $\Delta\text{MAE} -0.00986$

Unprotected opportunity cost

stronger comparator

12/12 budget violations

Risk-bounded SOH updates

before reference capacity is available